

hw4

Problem 1:

Problem 1. Consider polynomial least squares fitting (see your notes or Example 11.2 in the text) with $m = 100$, $n = 15$, points $t_i = (i - 1)/(m - 1)$, $i = 1, \dots, m$, and components of $\vec{b}$ given by $b_i = \exp(\sin(4 * t_i))$, $i = 1, \dots, m$.

(a) Show theoretically that the corresponding matrix $A \in R^{m \times n}$ has rank n . **Hint:** Use the fact that any nonzero polynomial of degree $\leq n - 1$ has at most $n - 1$ distinct roots.

(b) Use MATLAB's "linspace" to set up the column vector pts containing the points $\{t_i\}_{i=1}^m$ and then set up the matrix A using $A(:, j) = pts.^j$ for $j = 1, \dots, n$. Calculate and print $x(n)$ (the coefficient c_{n-1} in $p(t) = c_0 + c_1 t + \dots + c_{n-1} t^{n-1}$) by the following 4 methods:

(i) Form and then solve the system of normal equations using MATLAB's "chol" and your own functions " $\vec{x} = \text{forsub}(L, \vec{b})$ " and " $\vec{x} = \text{bacsub}(U, \vec{b})$ " that perform respectively forward substitution for a lower triangular L and backward substitution for an upper triangular U .

(ii) Use Matlab's " $[Q, R] = \text{qr}(A, 0)$ " (to obtain a reduced QR factorization of A) and your own function " $\vec{x} = \text{bacsub}(U, \vec{b})$ ".

(iii) Use a reduced SVD of A obtained by MATLAB's " $\text{svd}(A, 0)$ ".

(iv) Use MATLAB'S " $x = A \backslash b$ " which, for $A \in R^{m \times n}$ with $m > n$, computes the least squares solution making use of a QR factorization of A obtained by Householder transformations with column pivoting.

Remember to print $x(n)$ **only**. **Do not print** A , $\vec{b}$, or any intermediate matrices. Using the exact value $x(n) = 2.006787453080206 \dots e+03$, determine the number of correct significant digits in $x(n)$ computed by the methods (i)–(iv).

To make sure that your code (i) works correctly, test (i) and (ii) for $m = 10$ and $n = 5$. You should obtain similar values of $x(n)$ which are to be printed and submitted.

a)

A is a vandermonde matrix where row i is $[1 \ t_i \ \dots \ t_i^{n-1}]$ so then $A_{ij} = t_i^{j-1}$. To show, $\text{rank}(A)=n$, we WTS $Ax=0 \implies x$ is 0 vector

Let $Ax=y$. Then $y_i = \sum_{j=1}^n x_j A_{ij} = \sum_{j=1}^n x_j t_i^{j-1}$

We can rewrite this as $f(t) = x_1 + x_2 t + \dots + x_n t^{n-1} = 0$

If x_j is not all 0s, then the max degree of the polynomial is $n-1$ so the max amount of roots is $n-1$. However, we know that $f(t_i) = 0 \ \forall i = 1 \dots m$. since, $m > n$, this means there are more 0s than roots which is only possible if $f(t)=0$ so all x_j must be 0, so x is the 0 vector

b)

```
function [A,b] = make_matrix(m, n)
    pts = linspace(0,1, m)';
    A= ones(m, n);
```

```

    for j = 2:n
        A(:, j) = pts .* A(:, j-1);
    end

    b = exp(sin(4 * pts));

end

function x = forsub(L, b)
    n = length(b);
    x = zeros(n, 1);
    x(1) = b(1) / L(1,1);
    for i = 2:n
        x(i) = (b(i) - L(i, 1:i-1) * x(1:i-1)) / L(i,i);
    end
end

function x = bacsub(U, b)
    n = length(b);
    x = zeros(n, 1);
    x(n) = b(n) / U(n,n);
    for i = n-1:-1:1
        x(i) = (b(i) - U(i, i+1:n) * x(i+1:n)) / U(i,i);
    end
end

exact_xn = 2.006787453080206e+03;

% Part 1
fprintf("Part 1) \n")

% Test
[A, b] = make_matrix(10, 5);
lhs = A' * A;
rhs = A' * b;
L = chol(lhs, "lower");
U = L';
y = forsub(L, rhs);
x = bacsub(U, y);
fprintf("Test case: \n")
xn = x(5);
disp(xn);

try
    [A, b] = make_matrix(100, 15);
    lhs = A' * A;
    rhs = A' * b;

```

```

    L = chol(lhs, "lower");
    U = L';
    y = forsub(L, rhs);
    x = bacsu(U, y);
    fprintf("Actual: \n")
    xn = x(15);
    disp(xn);
    sig_figs = -log10(abs(xn - exact_xn) / abs(exact_xn));
    fprintf("Sig figs: %.2f\n", sig_figs);
catch
    fprintf("Cannot run Chol because of conditioning/positive definite\n")
end
% Part 2
fprintf("Part 2) \n")
[A, b] = make_matrix(10, 5);
[Q, R] = qr(A, 0);
rhs = Q' * b;
x = bacsu(R, rhs);
fprintf("Test case: \n");
xn = x(5);
disp(xn);

[A, b] = make_matrix(100, 15);
[Q, R] = qr(A, 0);
rhs = Q' * b;
x = bacsu(R, rhs);
fprintf("Actual: \n");
xn = x(15);
disp(xn);
sig_figs = -log10(abs(xn - exact_xn) / abs(exact_xn));
fprintf("Sig figs: %.2f\n", sig_figs);

fprintf("Part 3) \n")
[A, b] = make_matrix(100, 15);
[U, S, V] = svd(A, 0);
x = V * (S \ (U' * b));
fprintf("Actual: \n");
xn = x(15);
disp(xn);
sig_figs = -log10(abs(xn - exact_xn) / abs(exact_xn));
fprintf("Sig figs: %.2f\n", sig_figs);

% Part 4
fprintf("Part 4) \n")
[A, b] = make_matrix(100, 15);
x = A \ b;

```

```
fprintf("Actual: \n");  
xn = x(15);  
disp(xn);  
sig_figs = -log10(abs(xn - exact_xn) / abs(exact_xn));  
fprintf("Sig figs: %.2f\n", sig_figs);
```

Output:

```
Part 1)  
Test case:  
    22.8152  
  
Cannot run Chol because of conditioning/positive definite  
Part 2)  
Test case:  
    22.8152  
  
Actual:  
    2.0068e+03  
  
Sig figs: 6.68  
Part 3)  
Actual:  
    2.0068e+03  
  
Sig figs: 6.68  
Part 4)  
Actual:  
    2.0068e+03  
  
Sig figs: 6.57
```

Problem 2

$$\begin{aligned}
 A(x + \delta x) &= b + \delta b \\
 Ax + A\delta x &= b + \delta b \quad \text{Since } Ax = b \Rightarrow A\delta x = \delta b \Rightarrow \delta x = A^{-1}\delta b \\
 \|\delta x\| &= \|A^{-1}\delta b\| \leq \|A^{-1}\| \|\delta b\| \quad \text{since by defn} \\
 \text{Since } \|A^{-1}\| &= \max_{\|z\|=1} \|A^{-1}z\| \exists z \text{ st. } \|A^{-1}\| = \frac{\|A^{-1}z\|}{\|z\|} \\
 \text{let } \delta b &= z \text{ so } \|A^{-1}\delta b\| = \|A^{-1}\| \|\delta b\| \\
 \text{we can use this same property to let } \|b\| &= \|Ax\| = \|A\| \|x\| \\
 \text{so } \|x\| &= \frac{\|b\|}{\|A\|} \\
 \text{Then } \| \delta x \| &= \|A^{-1}\| \|\delta b\|, \quad \frac{\|\delta x\|}{\|x\|} = \frac{\|A^{-1}\| \|\delta b\|}{\|x\|} = \frac{\|A^{-1}\| \|\delta b\|}{\|b\| / \|A\|} \\
 &= \frac{\|A^{-1}\| \|\delta b\|}{\|b\|} \cdot \|A\| = \|A\| \|A^{-1}\| \frac{\|\delta b\|}{\|b\|} = K(A) \frac{\|\delta b\|}{\|b\|} \\
 \text{so } \frac{\|\delta x\|}{\|x\|} &= K(A) \frac{\|\delta b\|}{\|b\|}
 \end{aligned}$$

Problem 3

$$f(x) = \frac{1 - \cos(x)}{x^2} = \frac{(1 - \cos(x))(1 + \cos(x))}{x^2(1 + \cos(x))} = \frac{1 - \cos^2(x)}{x^2(1 + \cos(x))} = \frac{\sin^2(x)}{x^2(1 + \cos(x))}, \text{ this is stable because no catastrophic cancellation}$$

$$K(x) = \frac{f'(x)}{f(x)} = \left| \frac{xf'(x)}{f(x)} \right|$$

$$f(x) = \cos x = \left(1 - \frac{x^2}{2!} + \frac{x^4}{4!} - \dots \right) / x^2$$

$$f'(x) = \left(\frac{x^2}{2} - \frac{x^4}{4!} + \frac{x^6}{6!} - \dots \right) / x^2$$

$$f(x) = \frac{1}{2} - \frac{x^2}{4!} + \frac{x^4}{6!} - \dots$$

$$f'(x) = -\frac{2x}{4!} + \frac{4x^3}{6!} - \dots$$

$$K(x) \approx \left| \frac{x \left(-\frac{2x}{4!} + \frac{4x^3}{6!} + \dots \right)}{\frac{1}{2} - \frac{x^2}{4!} + \dots} \right| \quad \text{we remove the } \dots \text{ b/c}$$

they contribute minimally

$$\approx \left| \frac{-\frac{2x^2}{4!} + \frac{4x^4}{6!}}{\frac{1}{2} - \frac{x^2}{4!}} \right| \approx \left| \frac{-\frac{x^2}{12}}{\frac{12-x^2}{24}} \right| = \left| \frac{-x^2}{12} \cdot \frac{24}{12-x^2} \right| =$$

$$\left| \frac{24x^2}{12(12-x^2)} \right| \quad \lim_{x \rightarrow 0} K(x) = 0 \text{ so stable}$$

at $x=0$

Problem 4

$$f(x) = (1+x)^{\frac{1}{3}} - 1$$

Know $a^3 - b^3 = (a - b)(a^2 + ab + b^2)$

so $a - b = (a^3 - b^3)/(a^2 + ab + b^2)$

Let $a = (1 + x)^{\frac{1}{3}}$ and $b=1$

so $f(x) = x/((1 + x)^{\frac{2}{3}} + (1 + x)^{\frac{1}{3}} + 1)$ Again, no cancellation

$K(x) = |xf'(x)/f(x)|$, $f'(x) = \frac{1}{3}(1 + x)^{-\frac{2}{3}}$

so $K(x) = x\frac{1}{3}(1 + x)^{-\frac{2}{3}}/((1 + x)^{\frac{1}{3}} - 1)$

Using taylor expansion around $x=0$, we can approx $(1 + x)^{\frac{1}{3}} \approx 1 + \frac{1}{3}x$

To get $K(x) \approx x\frac{1}{3}(1 + x)^{-\frac{2}{3}}/(\frac{1}{3}x) = (1 + x)^{-\frac{2}{3}}$

and limit of that as x approaches 0 is 1 so it is stable

Problem 5

Code

```
f = @(x) sqrt(x+1) - 1;
g = @(x) x / (sqrt(x+1) + 1);
for i= 1:5
    x = 100^i
    unstable = f(x);
    stable = g(x);
    fprintf("Unstable: %e\n", unstable);
    fprintf("Stable: %e\n", stable);
end

unstable = f(10^-16);
stable = g(10^-16);
fprintf("Unstable for x -> 0: %e\n", unstable);
fprintf("Stable for x -> 0: %e\n", stable);
```

Output:

```
x =

    100

Unstable: 9.049876e+00
Stable: 9.049876e+00

x =

    10000
```

Unstable: 9.900500e+01

Stable: 9.900500e+01

x =

1000000

Unstable: 9.990005e+02

Stable: 9.990005e+02

x =

100000000

Unstable: 9.999000e+03

Stable: 9.999000e+03

x =

1.0000e+10

Unstable: 9.999900e+04

Stable: 9.999900e+04

Unstable for x -> 0: 0.000000e+00

Stable for x -> 0: 5e-17

Here we see that there was total cancellation for the unstable, we lost all precision.
For stable we retain the precision and have the answer.

problem 6

$w = (x+y)$ so $\tilde{w} = z(x+y)(1+u_1)(1+u_2)$ b/c 2 operations
 where $|u_1|, |u_2| \leq u$ so $\tilde{w} = w(1+u_1)(1+u_2)$

$$\tilde{w} = w + w(u_1 + u_2 + u_1 u_2) \Rightarrow \tilde{w} - w = w(u_1 + u_2 + u_1 u_2)$$

$$\Rightarrow \frac{\tilde{w} - w}{w} = u_1 + u_2 + u_1 u_2 \Rightarrow \left| \frac{\tilde{w} - w}{w} \right| = |u_1 + u_2 + u_1 u_2|$$

$$\text{since } |u_1|, |u_2| \leq |u|$$

$$\left| \frac{\tilde{w} - w}{w} \right| \leq 2u + u^2$$

~~and $\left| \frac{\tilde{w} - w}{w} \right| \leq 2u + u^2$ since $|u_1|, |u_2| \leq u$ so $\tilde{w} = w(1+u_1)(1+u_2)$~~

Let $\tilde{x} = x(1+u_1)$, $\tilde{y} = y(1+u_1)$, $\tilde{z} = z(1+u_2)$

$$w(\tilde{x}, \tilde{y}, \tilde{z}) = \tilde{z}(1+u_2)(\tilde{x}(1+u_1) + \tilde{y}(1+u_1)) = \tilde{z}(\tilde{x} + \tilde{y}) = z(x+y)(1+u_1)(1+u_2) = \tilde{w}(x, y, z)$$

So $w(\tilde{x}, \tilde{y}, \tilde{z}) = \tilde{w}(x, y, z)$ so

stably
 $\frac{\tilde{w} - w}{w} \leq 2u + u^2$